

MODULE 3 OF 6

Exploratory Data Analysis and the Statistics of Grade

Before grade can be estimated in space, it must be understood as a number

This module builds the statistical description of a grade dataset — its distribution, skew, outliers and hidden spatial bias — and ends at the single most important judgement in the workflow: the decision of the stationary domain.

LEARNING OBJECTIVES

- Compute and interpret the univariate statistics of a grade distribution.
- Explain why grades are lognormal and use the log-transform correctly.
- Recognise clustered sampling and correct the bias by declustering.
- Diagnose outliers and decide a defensible top-cut.
- Composite samples to a common support.
- State the stationarity assumption and define geological domains.

1 The univariate description of grade

Treat the n grades as realisations of a random variable Z . The **arithmetic mean** estimates the deposit's average grade:

$$\bar{z} = \frac{1}{n} \sum_{i=1}^n z_i \quad (3.1)$$

The **sample variance** uses the divisor $n - 1$ (Bessel's correction) for unbiasedness, and the **coefficient of variation** rescales the spread by the mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})^2, \quad \text{CV} = \frac{s}{\bar{z}} \quad (3.2-3.3)$$

A CV below 0.5 is a well-behaved deposit; above 1 signals an erratic, high-variance grade that will be hard to estimate. The **coefficient of skewness** is the third standardised moment:

$$\gamma_1 = \frac{\frac{1}{n} \sum_{i=1}^n (z_i - \bar{z})^3}{s^3} \quad (3.4)$$

Running these formulas on the 32 field-sample SnO₂ grades: $n = 32$, minimum 0.15%, maximum 8.84%, mean **2.27%**, median **1.50%**, standard deviation 2.21%, **CV 0.98**, skewness **+1.48**. The mean exceeds the median by 51%, the CV is almost 1, and skewness

is strongly positive. Every statistic tells the same story: a skewed, high-variance grade — which is how metal grades almost always behave.

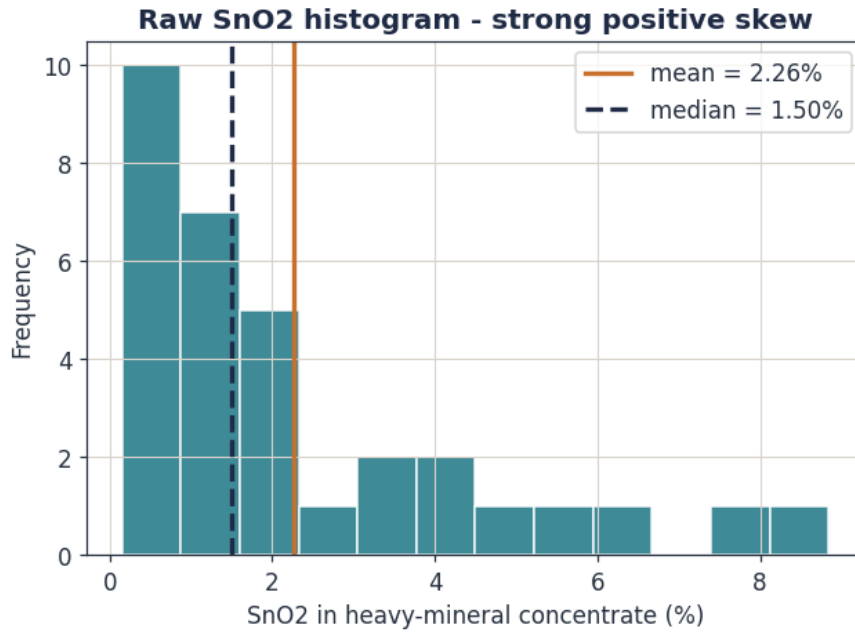


Figure 1 – Histogram of the 32 Toro SnO₂ grades. The long right tail drags the mean well above the median.

2 The lognormal model of grade

Grade is the product of many multiplicative geological processes, and when independent random factors multiply, their product tends towards a **lognormal** distribution: $Y = \ln Z \sim \mathcal{N}(\mu, \sigma^2)$. The lognormal model explains the mean-median gap exactly:

$$\text{median}(Z) = e^{\mu}, \quad \mathbb{E}[Z] = e^{\mu + \sigma^2/2}, \quad \text{Var}(Z) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2} \quad (3.6)$$

Because $\sigma^2/2 > 0$, the mean always exceeds the median. The diagnostic tools follow: the log-transform should symmetrise the histogram, and the lognormal probability plot should straighten the ordered log-grades. For Toro, the log-transform collapses skewness from +1.48 to -0.09 — the grade is, to a good approximation, lognormal.

$\mathbb{E}[e^Y] \neq e^{\mathbb{E}[Y]}$: exponentiating a kriged log-grade does not give the kriged grade, and the error is a systematic bias. The safe default (Module 5) is to estimate in original units with ordinary kriging after a sensible top-cut.

3 Clustered sampling and declustering

Geologists drill where they expect grade, so high-grade ground is over-sampled and the naive mean *overstates* the deposit. Declustering replaces the equal-weight mean with a weighted mean in which each weight reflects the area a sample represents:

$$\bar{z}_{dc} = \sum_{i=1}^n w_i z_i, \quad \sum_{i=1}^n w_i = 1 \quad (3.7)$$

In *cell declustering*, a grid is laid over the data; each occupied cell gets an equal share of weight, split among the samples inside it. The Toro sampling is markedly clustered — most of the 26 pits lie in the Bishiwai and Yadabongo drainages, with only a handful at Kadade — so the naive mean of 2.27% is biased towards Bishiwai.

4 Outliers and the top-cut decision

In a skewed dataset a few extreme values exert leverage far beyond their number: the single 8.84% Toro value contributes about 12% of the summed grade. **Top-cutting** resets values above a chosen cap to the cap. The cap is documented judgement, informed by the probability-plot break, the cumulative curve, and the metal-at-risk in the top one or two samples. Toro's 97.5th percentile is 7.87%, and a cap of 7–8% removes the leverage of the two highest values while discarding very little real metal. Whatever cap is chosen must be stated, justified and its effect on contained metal disclosed — an undisclosed top-cut is a reporting failure under every code.

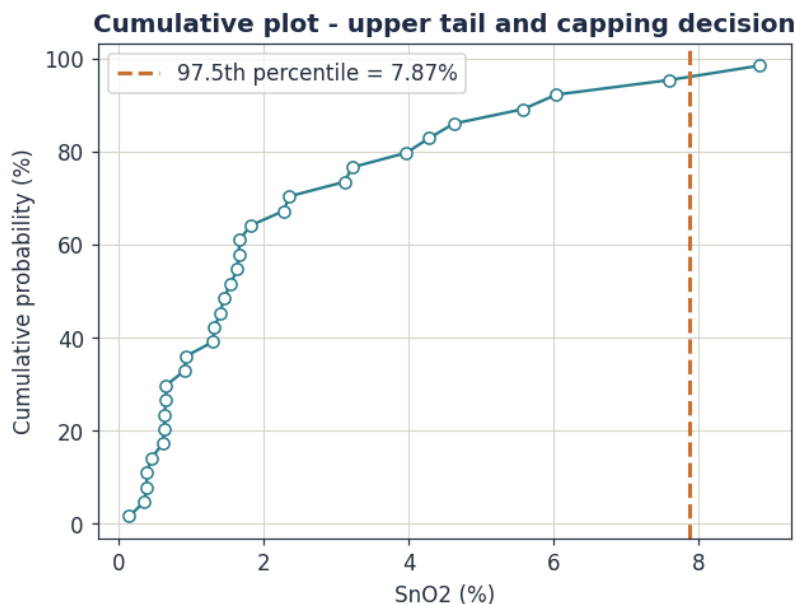


Figure 2 – Cumulative probability plot. The top two samples sit above a visible inflection near 7.9% SnO₂.

5 Compositing: putting samples on equal support

Statistics implicitly assume comparable data, but a 0.5 m drill interval and a 2 m interval are different **supports** (the size, shape and orientation a grade refers to). Compositing combines raw samples into regular intervals of common support, length-weighted:

$$z_{\text{comp}} = \frac{\sum_i \ell_i z_i}{\sum_i \ell_i} \quad (3.8)$$

Composited grades scatter less than the raw samples they are built from — the first appearance of the **support effect**, formalised by the volume-variance relationship in Module 6.

The Module 2 audit composited Toro layer samples with a simple mean only because the field logs do not consistently record sampled thickness. Equation (3.8) shows why that is a real compromise: without ℓ_i , a thin rich gravel stringer and a thick lean one are averaged as equals. Recording the sampled interval on every future pit is a prerequisite for a correct composite.

6 Domains and the decision of stationarity

Every estimation method assumes that the pooled data belong to one population — that grade behaves consistently across the volume estimated. That assumption is **stationarity**.

Stationarity is the assumption that the statistical properties of grade — mean, variance, spatial continuity — do not depend on absolute position within the volume considered. A **domain** is a volume of rock within which stationarity can reasonably be assumed: one mineralisation style, one grade population, one set of geological controls.

Stationarity is not provable; it is, in Journel’s phrase, a *decision*, defended by geological reasoning and supported by statistics: side-by-side histograms, box plots, and **contact analysis** (grade against distance across a proposed boundary — a step is a hard boundary, a gradient is a soft one).

If two genuinely different populations are pooled, the result is wrong everywhere. A high-grade vein pooled with its low-grade halo produces a variogram that describes neither and a kriging that bleeds vein grade into waste. At Toro the non-negotiable boundary is between the **alluvial** and **primary** targets: different rocks, different grade units (g/m³ vs % Sn), different geometries and different populations. They must be estimated as separate domains.

Within the alluvial target, each drainage (Bishiwai, Yadabongo, Kadade) is geologically a separate sediment-transport system. Present data cannot resolve whether they are one statistical population or three, so the conservative working decision is to carry them as sub-domains and revisit once drilling adds data.

7 Worked example: EDA on the Toro grades in Python

```
import numpy as np, pandas as pd
from scipy import stats
```

```
fs = pd.read_csv("data/toro_field_samples.csv")
z = fs["Sn02"].values

mean = z.mean()      # 2.27
median = np.median(z) # 1.50
sd = z.std(ddof=1)    # 2.21 (Bessel n-1)
cv = sd / mean        # 0.98
skew = stats.skew(z)  # +1.48

# lognormal test: log-transform should symmetrise
lz = np.log(z)
print(stats.skew(lz)) # -0.09 approximately symmetric

# metal at risk in the top two of 32 samples
top2 = np.sort(z)[-2:].sum() / z.sum() * 100
print(f"top 2 samples carry {top2:.0f}% of summed grade") # 23%
```

Two samples out of 32 — 6% of the data — carry 23% of the summed grade. That concentration of leverage is exactly what capping is designed to control, and exactly what the codes require you to disclose.

Module 3 of the Toro Resource Estimation course. Next: Module 4, Spatial Continuity and Variography.